

BUS713 Financial Analytics: Workshop 3

Portfolio Return, Risk-Adjusted Performance, and Diversification

Builds on Modules 1 and 2: return and risk, now combined into performance evaluation

Modules 1 and 2 measured return and risk separately. Module 3 puts them together: how a portfolio's return is actually built from its holdings, what that return is worth after inflation, and whether it was earned efficiently given the risk taken.

Task 1: Holding Period Return and Weighted Portfolio Return

Scenario: *You're a junior analyst preparing a client's annual portfolio review. The client holds several positions, and the client's spouse has been quoting only the best-performing one at dinner parties.*

Holding Period Return (HPR) combines capital gain and income received over one period, relative to the starting value: $HPR = (\text{Ending value} - \text{Beginning value} + \text{Income}) / \text{Beginning value}$. A portfolio's overall return is the weighted average of its individual holdings' returns, where each weight is that holding's share of total portfolio value.

1. Your client's portfolio was worth \$85,000 at the start of the year. By year-end it was worth \$91,800, after also paying out \$2,100 in dividends. Calculate the HPR and explain what it captures that a simple price return would miss.
2. The client holds three positions: 45% in a growth equity fund that returned 18%, 35% in a bond fund that returned 4%, and 20% in cash that returned 1%. Calculate the portfolio's weighted average return.
3. The client's spouse tells friends "our portfolio returned 18% last year," citing only the growth fund. Using your answer to problem 2, explain in 2 to 3 sentences why this is misleading.

Task 2: Arithmetic vs Geometric Returns

Scenario: *Your manager wants to state the client's typical annual return in a year-end letter and asks you to prepare the number.*

The arithmetic average simply averages a series of returns; it ignores compounding. The geometric mean return finds the constant annual rate that would produce the same final value, and is the more accurate measure of realised multi-period growth.

4. The client's portfolio returned +18%, -12%, +9%, and +6% over four consecutive years. Calculate the arithmetic average return and the geometric mean return.
5. Which figure is higher, and why does that follow from the sequence of gains and losses in problem 1?
6. Your manager wants to describe the portfolio's "typical annual return" in the client letter. Which average should you use, and what would happen to your credibility with the client if you used the other one instead?

Task 3: Nominal vs Real Returns

Scenario: *A retirement client is pleased with a strong nominal return during a year of unusually high inflation.*

A nominal return is measured in money terms only. A real return adjusts for inflation to measure the actual change in purchasing power. The Fisher equation gives the precise relationship: $(1 + \text{real return}) = (1 + \text{nominal return}) / (1 + \text{inflation rate})$.

7. The client's retirement portfolio earned a nominal return of 9% during a year when inflation ran at 6%. Use the Fisher equation to calculate the real return.
8. The client is pleased with the 9% nominal figure. Using your answer to problem 1, explain in 2 to 3 sentences what has actually happened to their purchasing power.
9. Recalculate the real return assuming inflation had instead been only 1%. Why can the same 9% nominal return represent very different outcomes for this client depending on the inflation environment?
10. Download S&P 500 price data since 2000 using `tq_get()` (ticker "`^GSPC`"), and Consumer Price Index (CPI) data for the same period using `tq_get()` with `get = "economic.data"`. Calculate the CAGR of the S&P 500 price level and the CAGR of the CPI over the full period, then use the Fisher equation to convert the nominal CAGR into a real, inflation-adjusted CAGR. What has the S&P 500's real annualised return been since 2000?

Task 4: Risk-Adjusted Performance: Sharpe and Treynor

Scenario: *Your firm is choosing between two external fund managers for a client allocation, and the client has only asked to see raw returns so far.*

The Sharpe ratio divides excess return (portfolio return minus the risk-free rate) by total risk, measured as standard deviation. The Treynor ratio divides the same excess return by systematic risk, measured as beta, and is more appropriate when portfolios are well diversified.

11. Manager X's portfolio returned 16% with a standard deviation of 14%. Manager Y's portfolio returned 13% with a standard deviation of 7%. Assume a risk-free rate of 4%. Calculate the Sharpe ratio for each manager.
12. Based on the Sharpe ratios, which manager delivered more return per unit of risk? Does this match the conclusion you'd reach by comparing raw returns alone?
13. Manager X's portfolio has a beta of 1.4 and Manager Y's has a beta of 0.9. Calculate the Treynor ratio for each, using the same returns and risk-free rate. Does the ranking change? If both portfolios are well diversified, which ratio should you actually rely on, and why?

Task 5: Downside Risk and the Sortino Ratio

Scenario: A risk-averse client tells you they don't mind upside swings, only losses, and asks why you keep reporting standard deviation as if both mattered equally.

Downside deviation measures variability only in returns that fall below a minimum acceptable return (MAR), ignoring upside moves entirely. The Sortino ratio divides excess return over the MAR by downside deviation.

14. A portfolio's monthly returns over the last eight months were 0.07, -0.02, 0.11, 0.03, -0.05, 0.09, 0.02, -0.01. Using an MAR of 3%, calculate the downside deviation (hint: $\min(0, \text{returns} - \text{mar})$, then take the square root of the mean of the squared result).
15. Calculate the Sortino ratio using the average return, the MAR, and the downside deviation from problem 1.
16. Calculate the portfolio's ordinary standard deviation and compare it to the downside deviation. Explain why the two numbers differ, and why this client would likely prefer the Sortino ratio over the Sharpe ratio for evaluating this portfolio.

Task 6: Systematic Risk, Diversification, and a Client Recommendation

Scenario: The same retiree client from Module 1 and 2, currently holding KO and PEP, asks whether adding more stocks would make their portfolio safer, and what beta actually means for them.

Unsystematic risk is firm-specific and can be diversified away by holding more, less-correlated assets. Systematic risk is economy-wide and cannot be diversified away; beta measures a portfolio's sensitivity to it.

17. Using KO and PEP as your example, explain why holding both instead of just one reduces unsystematic risk but not systematic risk. What kind of event would still affect both stocks at the same time?
18. Suppose KO has a beta of 0.6 and a semiconductor stock being considered for the portfolio has a beta of 1.8. If the market rises 8%, what is the expected change in each stock's return based on beta alone? What does adding the semiconductor stock imply for the portfolio's overall market sensitivity?
19. For this risk-averse retiree, would you prioritise (a) a high Sharpe ratio, (b) a low beta, or (c) a high Sortino ratio when selecting new investments? Justify your choice in 3 to 4 sentences, and note one limitation of the measure you chose.

Wrap-Up

You calculated Holding Period Return and weighted portfolio return, compared arithmetic and geometric averages, adjusted returns for inflation using the Fisher equation, evaluated performance using the Sharpe and Treynor ratios, measured downside risk with the Sortino ratio, and connected diversification and beta to a real client recommendation. Together with Modules 1 and 2, you now have a full return-risk-performance toolkit.